Name : Yashwin

Roll number : 2611006220007

## Analyse the features of Arm processors in designing application

1) Analyze the feature of Arm debug architecture

→ Features of Arm debug architecture

- Breakpoint : Stops program execution at a specific instruction for debugging
- Watchpoint : Monitors specific memory locations & stops execution when they are accessed or modified
- Debug Access port : Allows an external debugger to access the processor & its memory
- JTAG Interface : provides a connection blw the processor & external debugging tools
- Single stepping : Executes one instruction at a time to check prog behaviours
- Debug registers : store control & status information req for debugging
- Trace support : Records program execution to help identify errors & analyze performance

2) Analyze ARM7TDMI 3 Stage pipeline

-> ARM7TDMI uses a 3 stage pipeline to improve instruction execution speed. It allows multiple instructions do be processed simultaneously

The three stages are:

- Fetch : The processor fetches the instruction from memory
- Decode : The instruction is decoded, & the required operation is identified.
- Execute : The processor performs the operation & store the result

Working

- While one instruction is being executed the next instruction is decoded
- At the same time, another instruction is fetched
- This over lapping increases processor performance

3) Analyze the necessity of memory system

-> Necessity of memory system

A memory system is required in a computer to store instructions, data & results so that the CPU can acess them whenever needed

## Importance of memory system

- Data storage
- Fast access
- CPU performance
- Temporary storage
- Permanent storage
- Efficient processing
- Data transfer

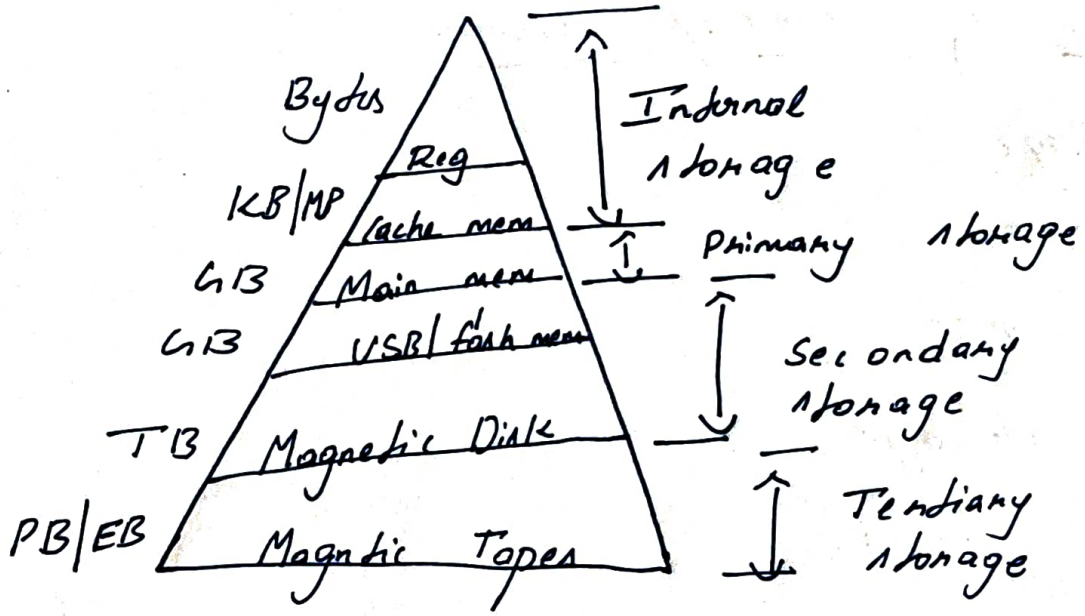
4) Analyze & explain memory hierarchy with necessary diagram

→ Memory hierarchy

Memory hierarchy is the arrangment of different types of memory in a computer system based on speed, cost & storage capacity.

It is designed to provide faster access to frequenty used data while maintaining large storage capacity at a reasonable cost

Memory hierarchy diagram




## levels of memory hierachy

1. Registers
2. Cache memory
3. Main memory
4. Secondary memory

5) Analyze & explain memory mangement system

## -) Memory management System

Memory management in the process of controlling & organizing the computer's main memory (RAM). It manages the allocation & deallocation of memory to different prog during execution

## Function of memory management :

1. Memory allocation
2. Memory deallocation

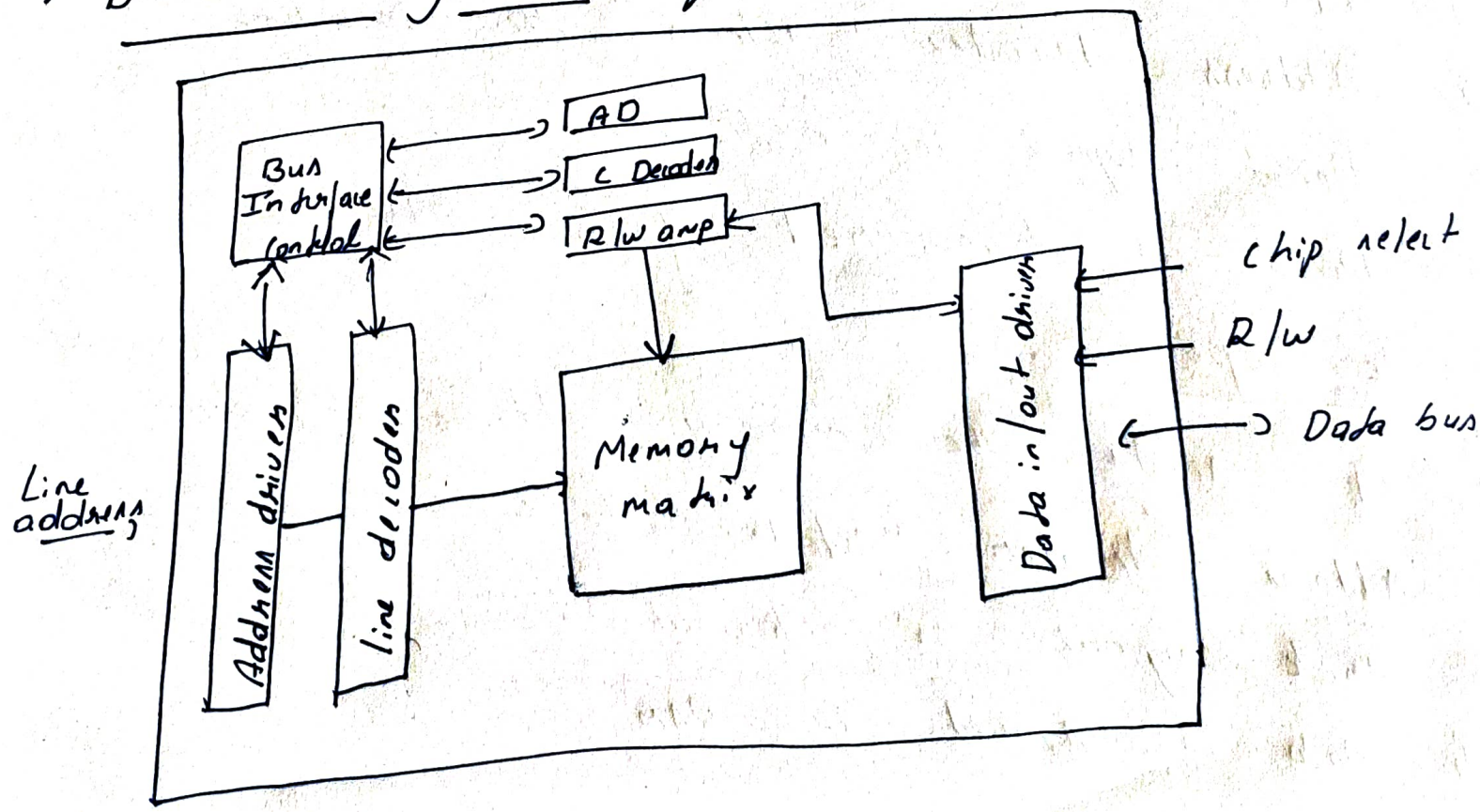
- Memory protection
- Address translation
- Memory sharing
- Memory utilization

## Types of memory mangement

- Contingous memory allocation
- Paging
- Segmentation
- Virtual memory

6) Analyze & explain block diagram of RAM chip & its operation with CPU

→ Block diagram of RAM chip



## Main Blocks

- **Address Bus** : carries the memory address from CPU
- **Address decoder** : Selects the req memory location
- **Memory array** : Stores data in binary form
- **Data bus** : Transfers data b/w RAM & CPU
- **Read/write** : Controls read & write operation
- **Chip select** : Enables the RAM chip

## Operation

### Read operation

1. CPU sends the memory address
2. Address decoder selects the location
3. Read signal is activated
4. Data is transferred from RAM to CPU

### Write operation

1. CPU sends the memory address & data
2. Address decoder selects the location
3. Write signal is activated
4. Data is stored in RAM

7) Analyze & explain various technology used in RAM

→ Various technolgies used in RAM

RAM is mainly classified into two types based on the technology

SRAM

- Stores data using ff circuits
- Does not require refreshing
- Very fast but expensive
- Used in cache memory

DRAM

- Stores data using capacitors & transistors
- Requires periodic refreshing
- Slower than SRAM but cheaper
- Used as main memory (RAM)

Types of DRAM

- SDRAM
- DDR RAM
  - DDR2, DDR3, DDR4, DDR5

8) Analyze & explain types of ROM chips & its operations with CPU

-) ROM (read only memory) is a non volati
- le memory used to store permanent data & instructions. It retains data even when power is off

## Types of ROM

### PROM (programmable ROM)

- Can be programmed only once
- Data cannot be erased

### EPROM (erasable PROM)

- Can be erased using ultraviolet light
- Can be reprogrammed

### EEPROM (electrically erasable PROM)

- Erased & reprogrammed electrically
- Data can be updated

## Flash ROM

- A type of EEPROM
- Erases & writes data in blocks
- Used in USB drives & memory cards

## operation of ROM with CPU

- CPU sends the required address through the address bus
- Address decoder selects the corresponding memory location
- Stored data is accessed from the memory array
- Data is transferred through the o/p buffer
- CPU receives the data through the data bus

9) Analyze & explain auxiliary & cache memory

→ Auxiliary memory is a non volatile secondary storage is used to store large amounts of data & prog permanently

### Features

1. Stores data permanently, even when power is off
2. Has large storage capacity
3. Slower than main memory
4. Cheaper than primary memory

Ex: HDD, SSD, CD

operation

- CPU requests data from secondary storage
- Required data is transferred to RAM
- CPU access the data from RAM for processing

Cache memory

Cache memory is a small, high speed memory located inside or near the CPU. It stores freq used data & instructions

Features

1. very fast memory
2. Smaller storage capacity than RAM
3. More expensive
4. Reduces CPU memory access time
5. stores freq accessed data

levels of cache

L1: fastest & smallest cache
L2: larger than L1 but slower
L3: larger than L2 & shared b/w CPU cores in many processors

## Operations

- CPU requests data
- Cache checks whether data is available
- If found, data is sent to CPU
- If not found, data is fetched from RAM & stored in cache

10) Analyze & explain temporal & spatial locality

→ Temporal & spatial locality

locality of ref means the CPU tends to access the same or nearby memory locations repeatedly

## Temporal locality

- Recently accessed data is likely to be accessed again
- Ex: A variable used repeatedly in a loop

## Spatial locality

- Nearby memory locations are likely to be accessed soon
- Ex: Accessing array elements one after another

11) Analyze & explain mapping functions & techniques with respect to memory make short

-> Mapping function determine how blocks of main memmory are placed into cache memory

. Direct Mapping

. Each memory block maps to only one fixed cache line
. Simple & fast
. May cause more conflicts

. Associative Mapping

. Any memory block can be placed in cache line
. Flexible but expensive
. Requires searching multiple cache lines

. Set Associative Mapping

. Cache is divided into sets
. Memory block maps to a particular set but can occupy any line within that set
. Combines advantages of direct & associative mapping

12) Analyze & explain virtual memory, paging, page replacement, page fault, segmented page map, TBL, & memory protection

## Virtual memory

- Allows prog larger than RAM to execute
- Uses secondary storage as an extension of RAM

## Paging

- Divides memory into fixed-size blocks
- Virtual blocks are called pages & physical blocks are called frames

## Page Replacement

- Replaces an existing page when RAM is full
Algorithms : FIFO, LRU & optimal

## Page fault

- Occurs when the required page is not in ram
- OS loads the page from secondary storage

## Segmented page Map

- Combines seg & paging
- Divides prog into segments & then into pages

## TLB

- Small, high speed memory in the MMU
- Stores recently used page table entries
- Speeds up address translation

## Memory protection

- Prevents unauthorized memory access
- Protects one process from another

13) Analyze & explain operation of nano memory with example

→ Nano memory refers to memory technology that uses nanoscale components to store & access data

## operation

- uses very small electronic components to store binary data
- Data is stored by changing the electrical state or resistance of memory cells

8. The memory controller reads & writes data
- Stored data can be accessed when required

Ex:-
- Nand flash memory

Advantages
- Small size
- High storage capacity
- Low power consumption

14) Consider length of instruction field 12 bits & address field is 4 bit each. I have 15 two address, 24 one address operation. Find out how many zero address operation we can have

-) Instruction field = 12 bits
Address field = 4 bits
Two address instructions = 15
one address instructions = 24
Find :- zero address instructions

## Two address instructions

Each instruction uses 2 address fields

Remaing instruction bits

$$12 - 4 - 4 = 4$$

Number of available combinations

$$2^4 = 16$$

15 are used

$$16 - 15 = 1$$

Step 2 : one address instructions

$$1 \times 2^4 = 16$$

24 one address instruction are given

$$(16 - 8) \times 16 = 128$$

Step 3 : zero address instruction

$$128 \times 2^4 = 2048$$

∴ 2048 zero address instructions